

ME 311P: Design Lab - 1

Course Code	ME 311P
Course Name	Design Lab - 1
L-T-P-C	0-0-2-1
Prerequisites	ME206 Mechanics of Solids
Intended for	B.Tech.
Distribution	Core
One-time approval	recommended by the 9th BoA (Oct 2015)
Replaced by	ME 206P by the 61st BoA (Jan 2026)

Course Contents

- Comparison of the properties of ductile and brittle materials.
- Effect of heat treatment on the mechanical properties.
- Tensile test to obtain stress strain curves, young's modulus, 0.2% proof stress, percentage elongation.
- Compression tests to compare the failure behaviour of the ductile and brittle materials.
- Shear test.
- Three point bend test for bending strength.
- Determination of Young's modulus of a material by bending test.
- Buckling load for columns with different end conditions.
- Evaluation of spring index and calculation of proof stress for compression spring.
- Impact tests and comparison of energies for ductile and brittle materials.
- Torsion tests to obtain modulus of rigidity and fracture behaviour of materials subjected to torsion.
- Hardness tests. Empirical relations of hardness and strength.
- Surface roughness tests to understand the different surface roughness obtained by various manufacturing processes.

Text Books

1. Timoshenko S. P., and Gere J. M., **Mechanics of Materials**, 2nd Edition, CBS Publishers, 2002.
2. Crandall S. H., Dahl N. C., and Lardner T. J., **An Introduction to the Mechanics of Solids**, 2nd Edition., McGraw-Hill, 1999
3. Hearn E. J., **Mechanics of Materials**, 3rd Edition, Pergamon, 2003.
4. Higdon A., Ohslen E. H., Stiles W. B., Weese J. A., and Riley W. F., **Mechanics of Materials**, John Wiley & Sons, 1989
5. Popov E. P., Nagarajan S., and Lu Z. A., **Mechanics of Materials**, 2nd Edition, Prentice-Hall of India, 2002.

References

1. Robert Cook, **Advanced Mechanics of Materials**
2. Alexander Blake, **Practical Stress Analysis in Engineering Design**
3. J. P. Den Hartog, **Advanced Strength of Materials**